6 A hot drink is cooling. The temperature of the drink at time t minutes is T°C.

The rate of decrease in temperature of the drink is proportional to $(T-20)$.

(a) Write down a differential equation to describe the temperature of the drink as a function of time. **[2]**

(b) When $t = 0$, the temperature of the drink is 90 °C and the temperature is decreasing at a rate of 4.9 °C per minute.

Determine how long it takes for the drink to cool from 90 °C to 40 °C. **[6]**